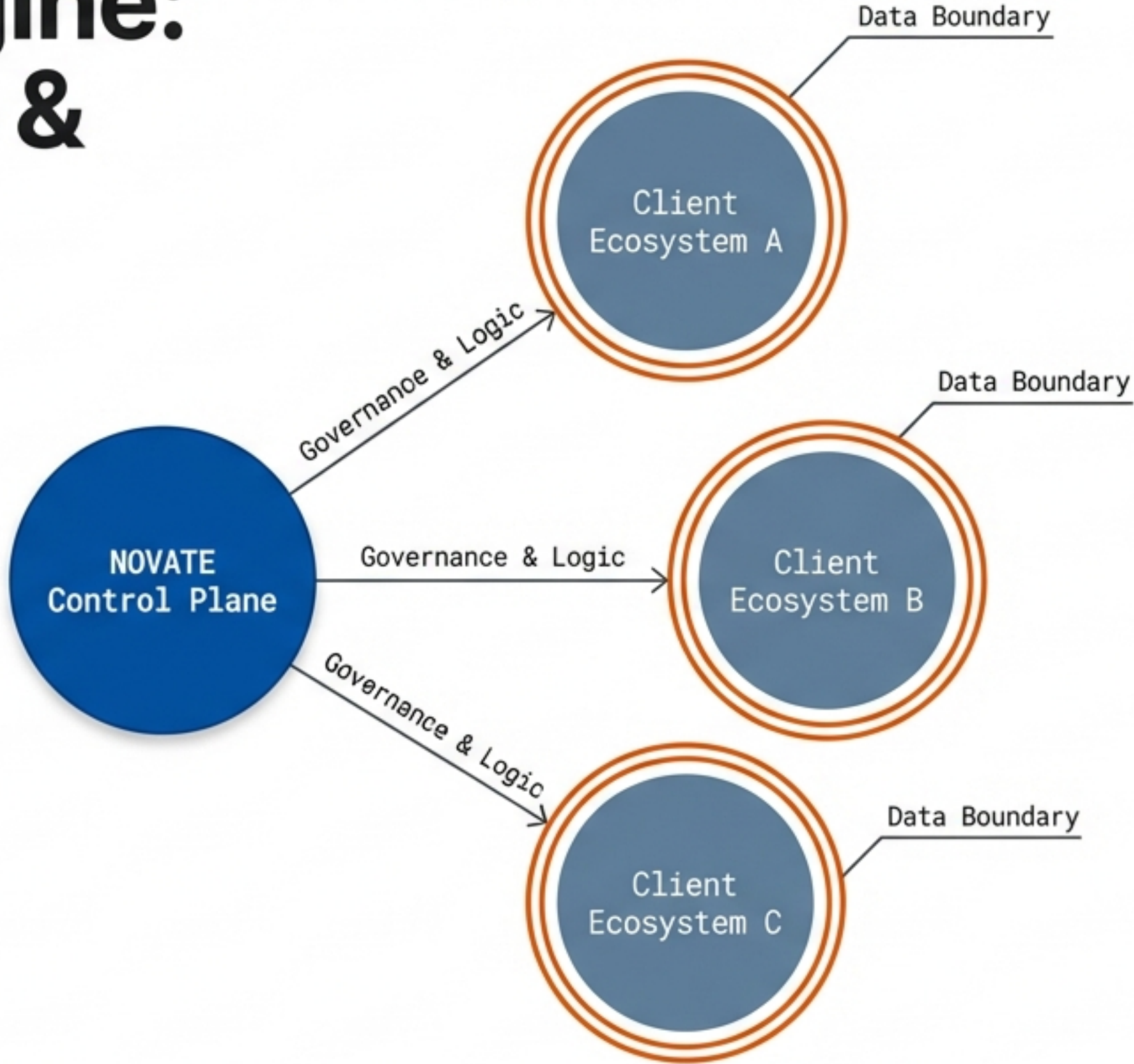


Sentra's Innovation Engine: Reference Architecture & Operational Strategy

A Multi-Tenant Control Plane
for Defensible Innovation.



OPERATOR MODEL
NOVATE runs the engine;
clients own the outcomes.

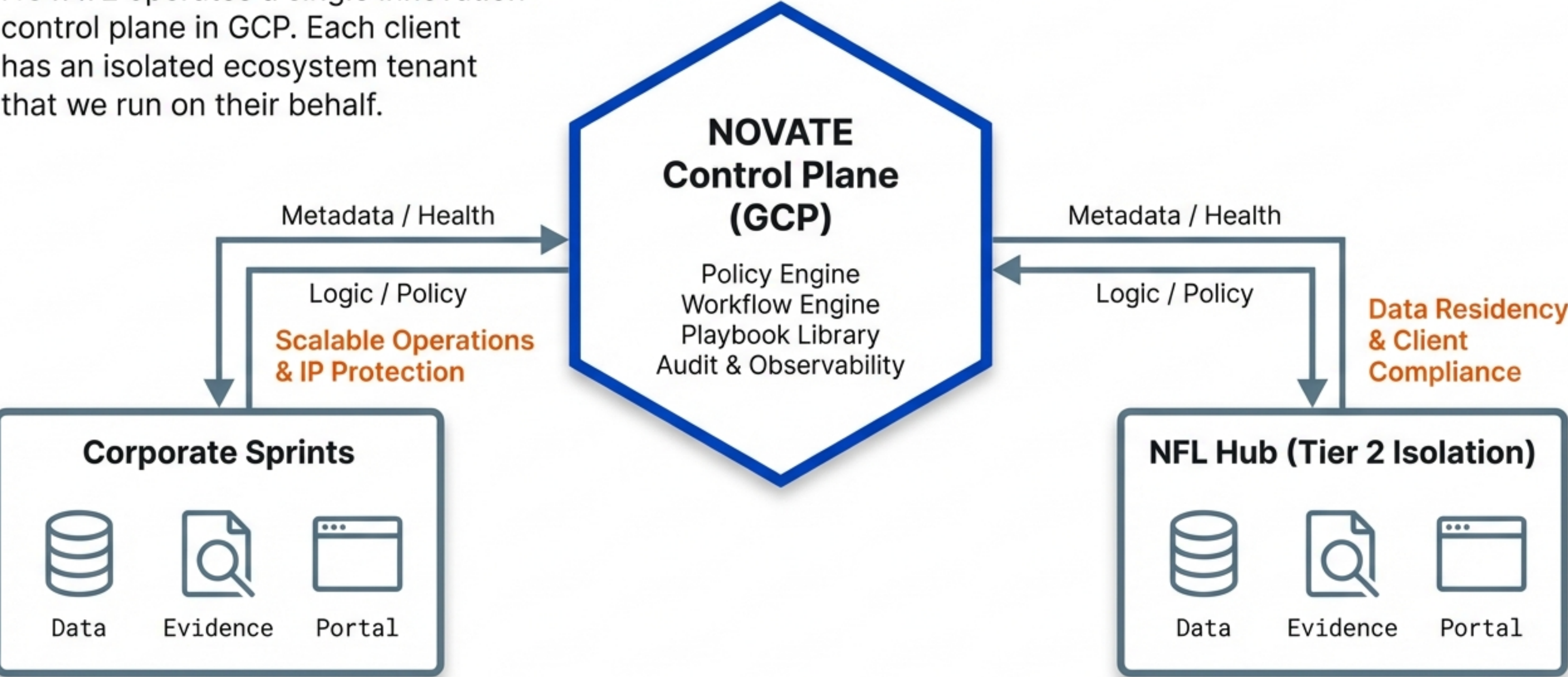
ISOLATION
Enterprise-grade tenant data
boundaries.

INTEGRATION
Connects to multi-cloud
client environments.

STANDARDIZATION
Powered by NIN (NOVATE
Innovation Network) standards.

One Control Plane. Many Ecosystems.

NOVATE operates a single innovation control plane in GCP. Each client has an isolated ecosystem tenant that we run on their behalf.



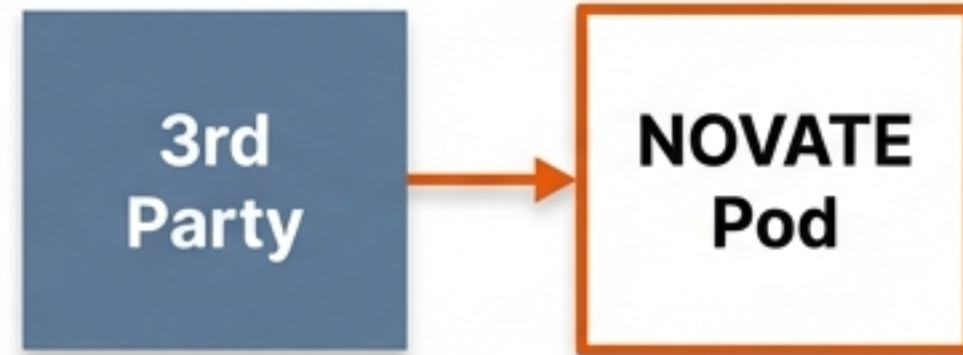
A Single Engine Supporting Three Innovation Modes

Mode 1: NOVATE-Hosted Community / Research Focus



- **Tenant:** NOVATE Foundation
- **Data:** Stored in NOVATE GCP
- **Portal:** Native Sentrais Participant Portal
- **Use Case:** Public hackathons, open innovation

Mode 2: Managed Innovation Client Programs



- **Tenant:** Client-specific isolated pod
- **Intake:** Native or 3rd Party (Submittable/Typeform)
- **Governance:** System-of-record is Sentrais
- **Use Case:** Corporate Challenges

Mode 3: White-Label Enterprise NFL Tier / Critical R&D



- **Tenant:** Hard Isolation (Tier 2 / Dedicated DB)
- **Controls:** Strict COI, IDS gating, Audit trails
- **IP:** NOVATE Playbooks deployed as templates
- **Use Case:** Internal R&D, IP Generation

The Architectural Stack: Core, Tenant, and Integration

Layer 1: NOVATE Control Plane (The Operator)

Context: Hosted in NOVATE GCP



Policy Engine
(Permissions/Gating)



Workflow Engine
(Stage Gates)



Registries
(Playbooks/Rubrics)



**Observability &
Governance Ops**
(Logs/Audit)

Layer 2: Client Ecosystem Pods (The Tenant)

Context: Tenant-Isolated Resources

Tenant Data Boundary 
(Encryption Keys/DB)



Ecosystem Objects
(Challenges/Submissions)



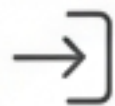
Client Portals
(Exec/Reviewer Views)



Branding Overlays
(Themes/Logos)

Layer 3: Channel & Integration Layer

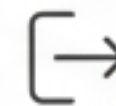
Context: Adapters & Connectors



Intake Adapters
(Native / 3rd Party)



Client Systems
(SSO, File Stores)

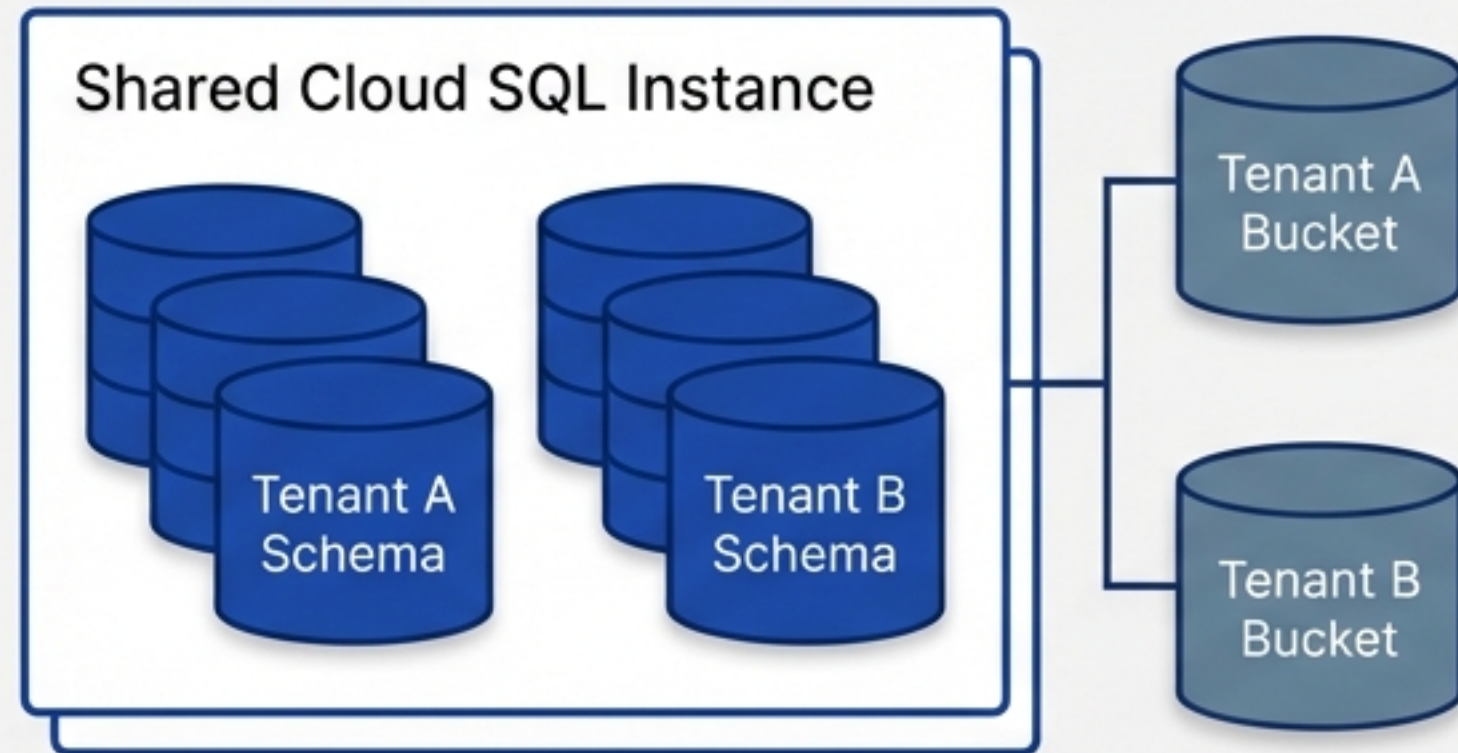


Data Export Connectors
(APIs, Webhooks)

Tenant Isolation & Data Security Strategy

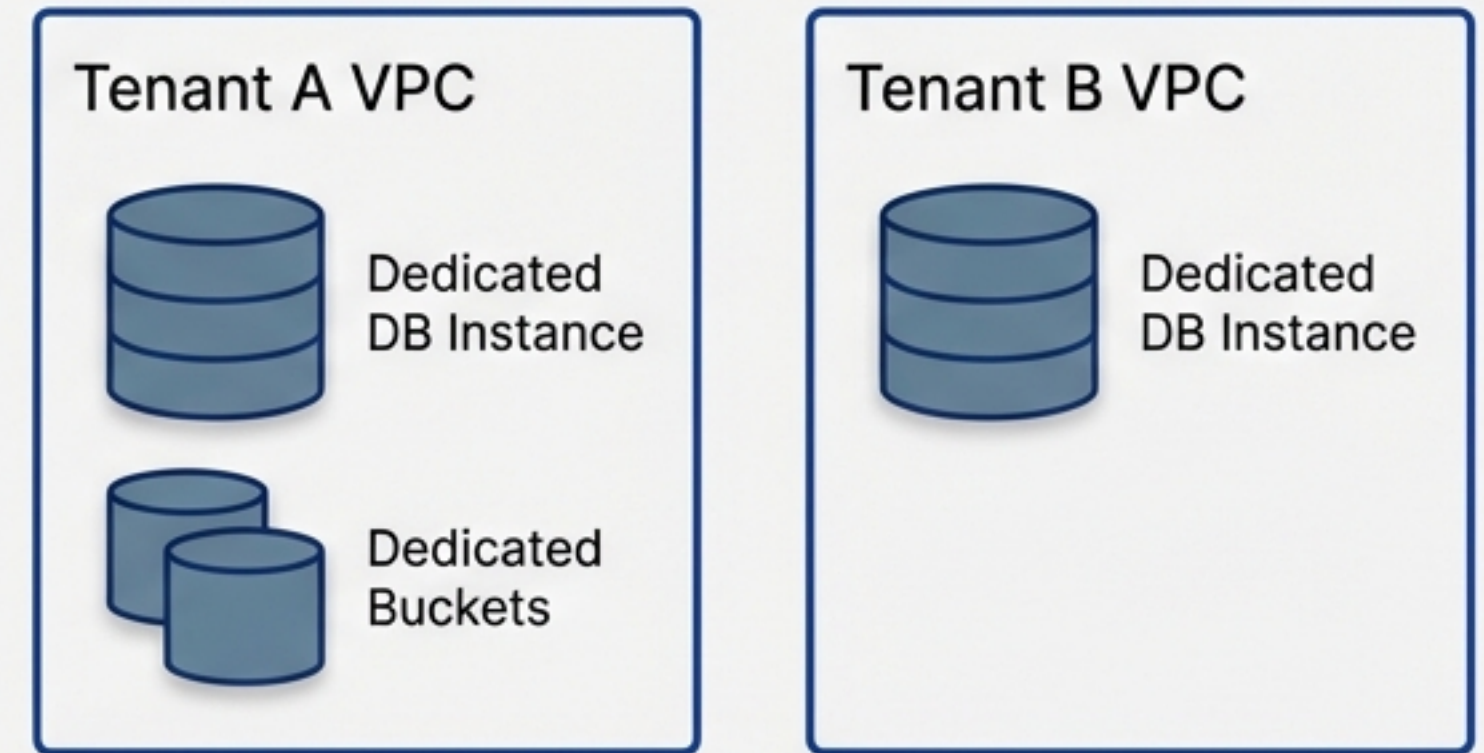
Control plane is shared; data plane is tenant-isolated.

Tier 1 (Standard Isolation)



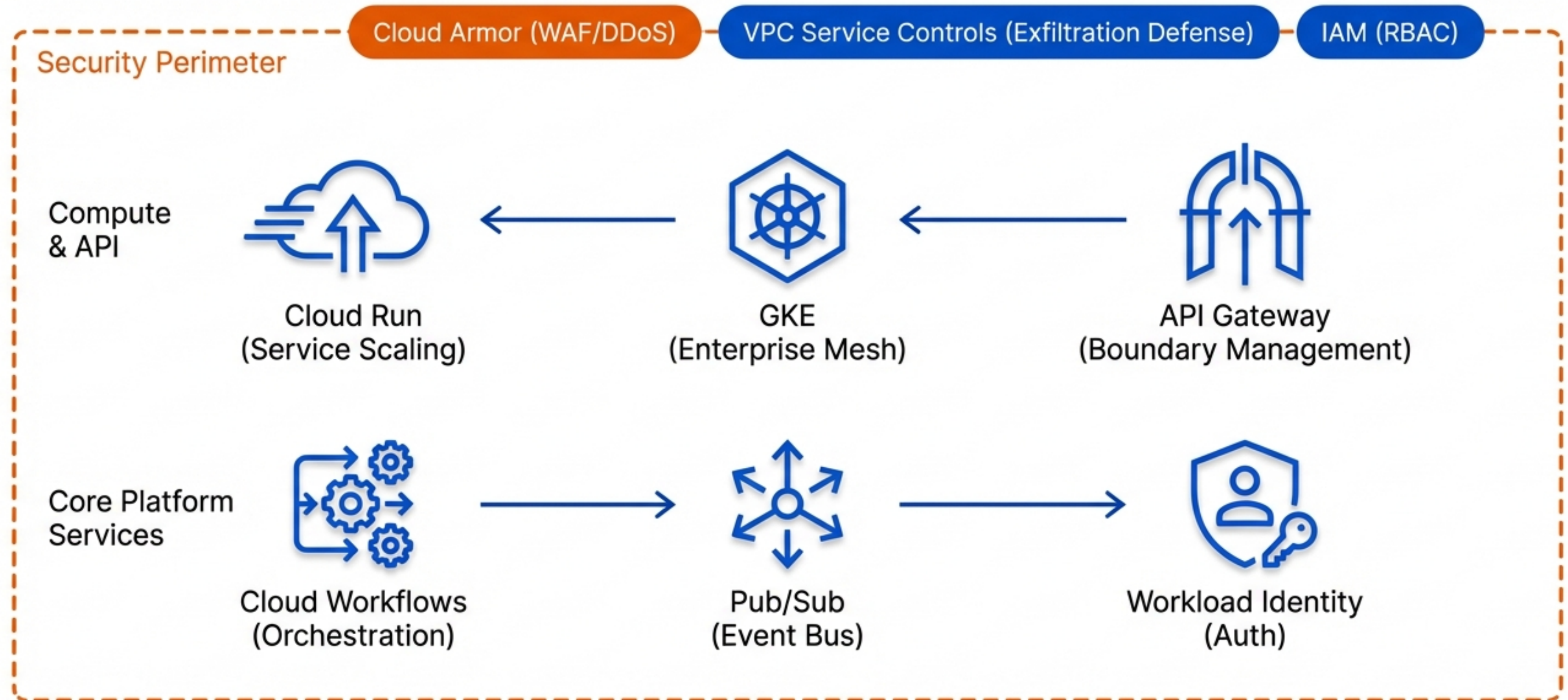
- Schema-per-tenant isolation
- Shared Compute Resources
- CMEK Support

Tier 2 (NFL / Critical Isolation)

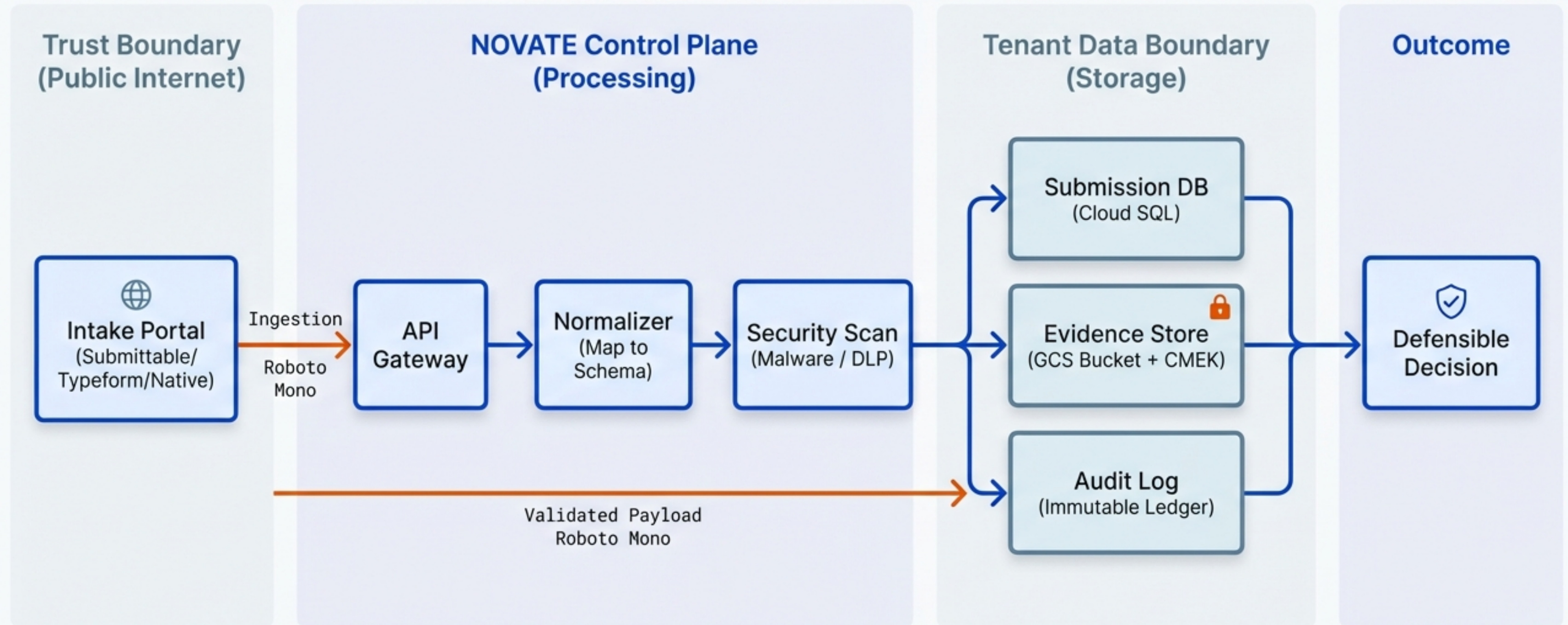


- Dedicated Database Instances
- Separate VPCs / Projects
- Strict Network Segmentation

GCP Reference Stack & Service Topology



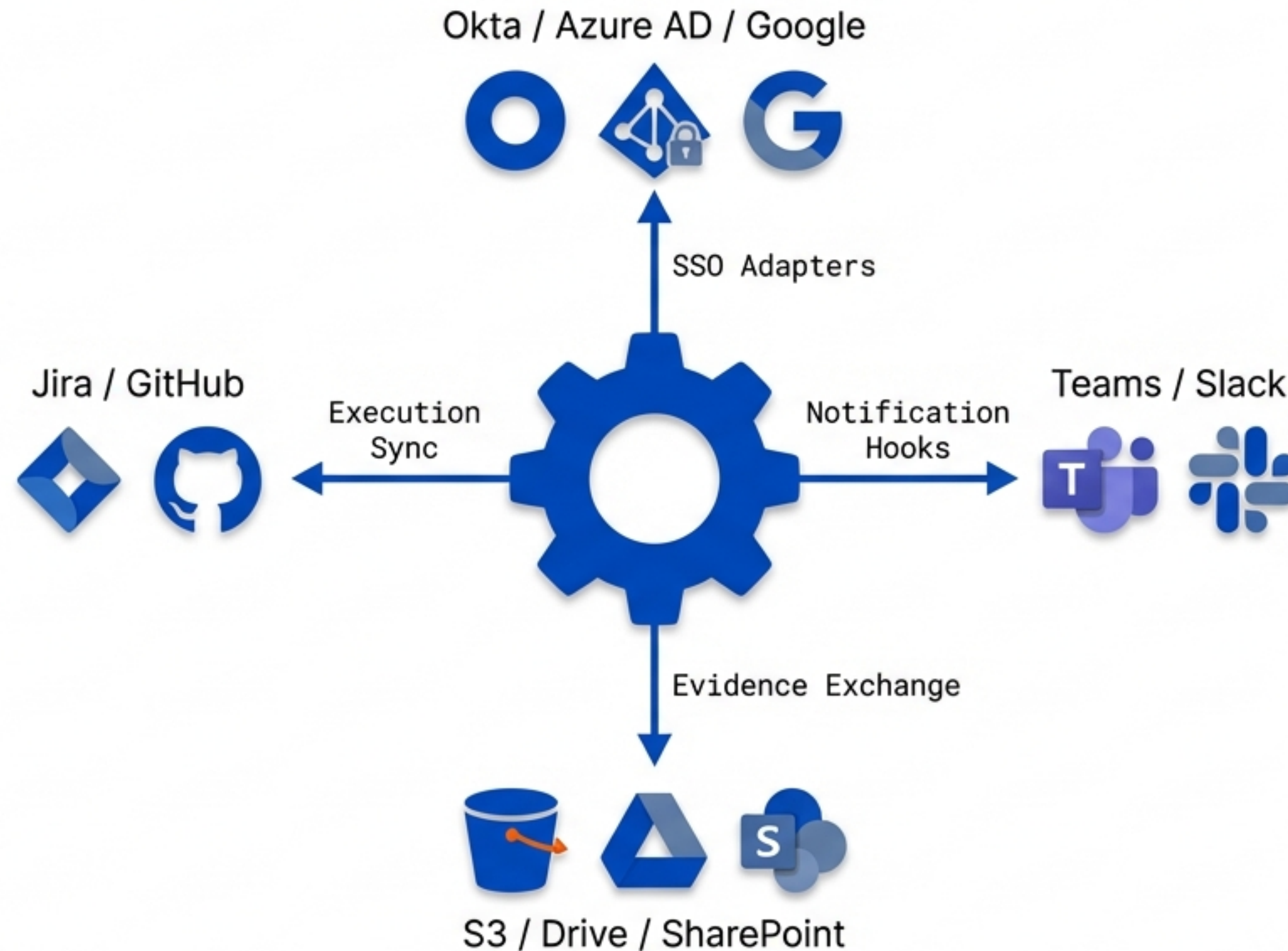
Data Flow Architecture: From Intake to Defensibility



Security Note: Third-party intake is merely a channel. Sentrais is the system of record. All external data crosses a trust boundary before storage.

Multi-Cloud Integration Strategy

Engine in GCP, Integrate Everywhere.



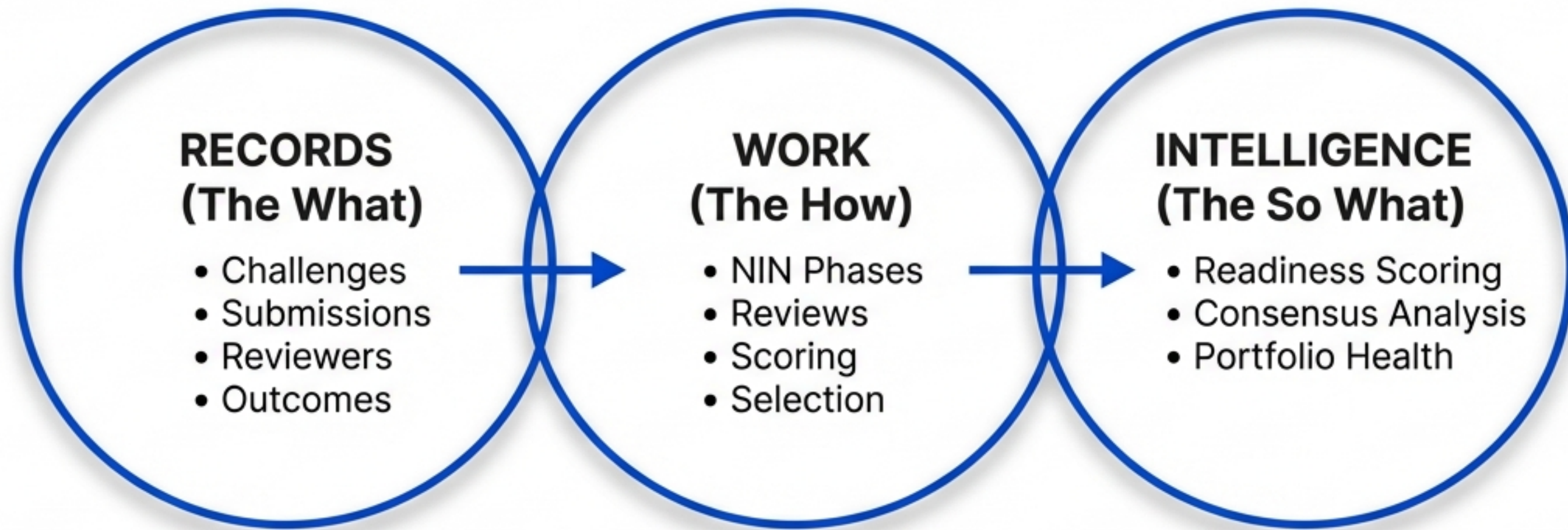
Philosophy: We do not move the engine; we move the data via connectors.

This solves the '**Shadow IT**' problem by centralizing governance while preserving tool flexibility.

Innovation Orchestration Standards (NIN)

Innovation feels disciplined, not experimental.

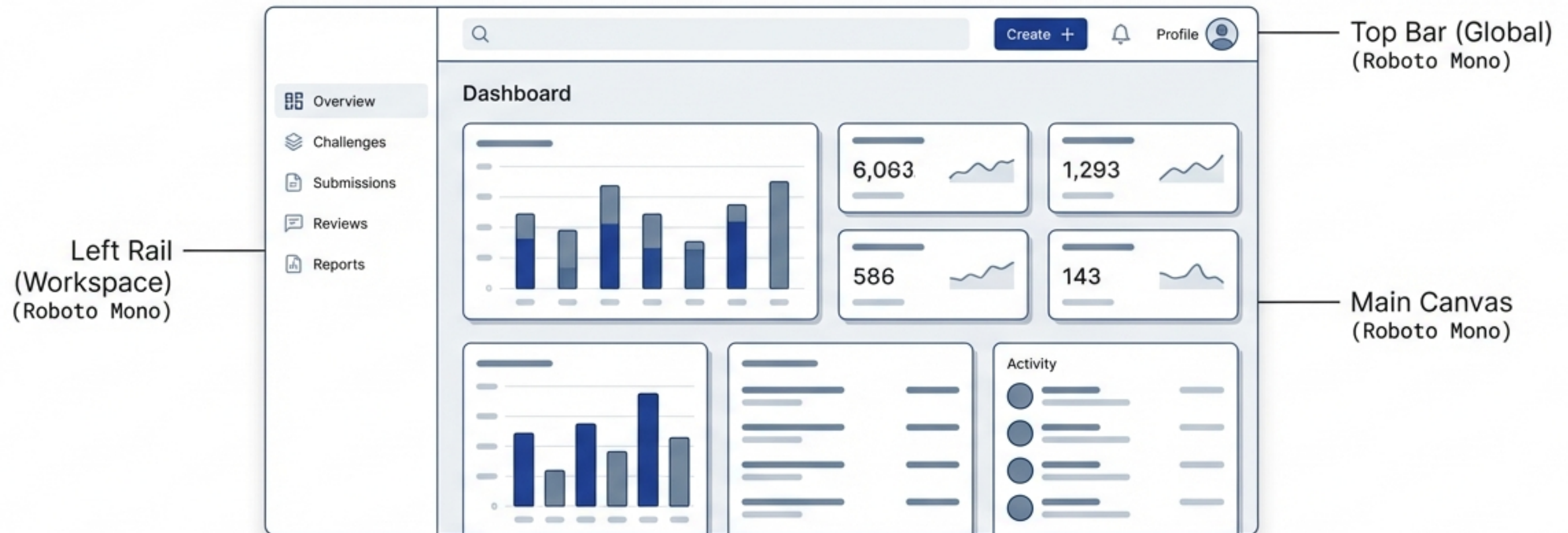
Problem-to-Outcome Pattern



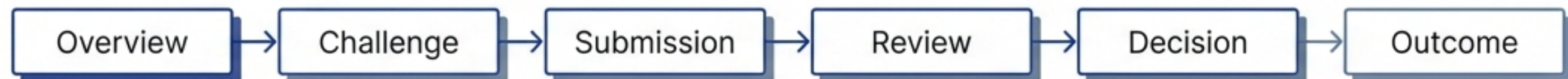
Workflow
Primitives



The Design System: Layout & Navigation



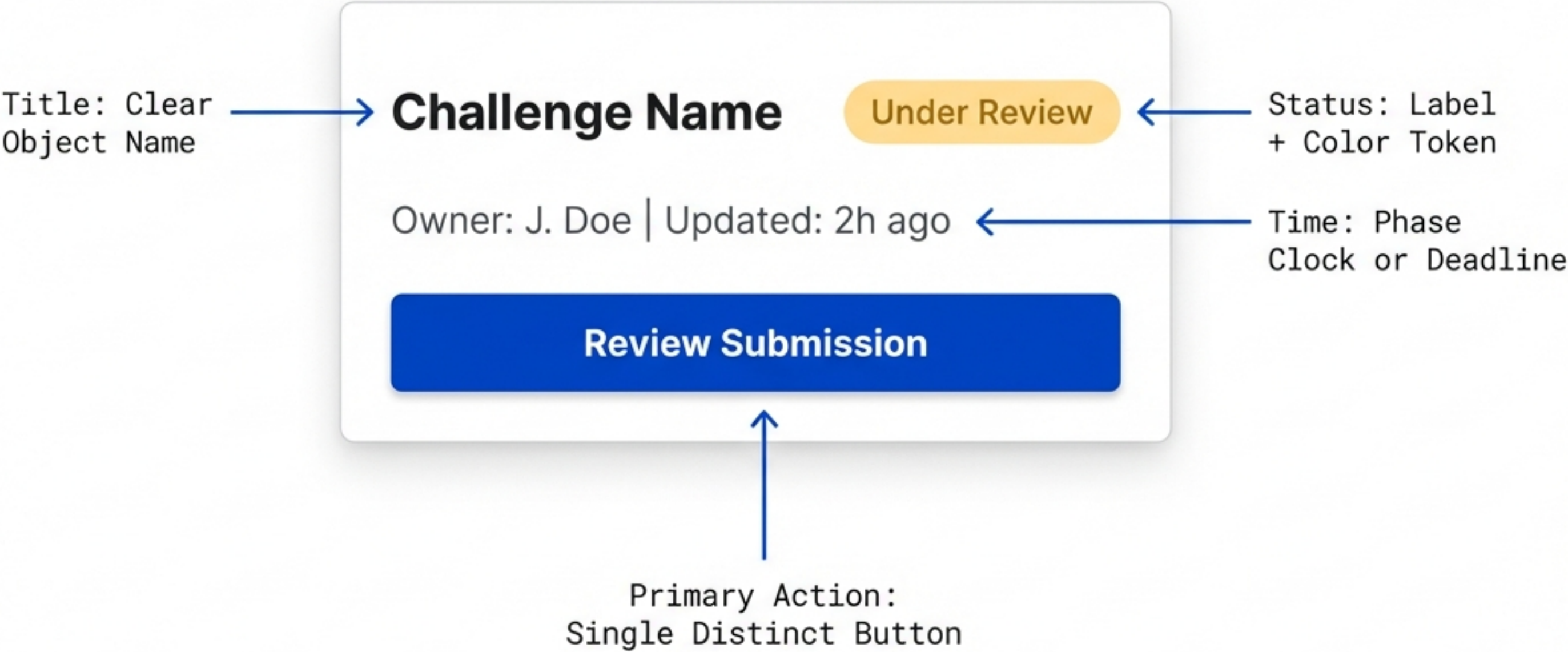
The Canonical Journey (Inter Bold)



Philosophy: Calm in complexity. Uses standard design tokens to ensure consistency with Operations.
Inter Regular, Roboto Mono

The Card System: Scannability & Action

Understandable and actionable in under 5 seconds.



Required Card Types

- Challenge
- Submission
- Review
- Score
- Insight
- Evidence

Governance & Permissions Model

	Read / Submit	Score / Comment	Manage Gates	Evidence / Logs
Observer				
Reviewer				
Operator				
Auditor				

Context Awareness:

Permissions change dynamically based on Phase (early vs. late) and Sensitivity (blind reviews).

Security Principle: Least privilege by default. All overrides generate audit log entries.

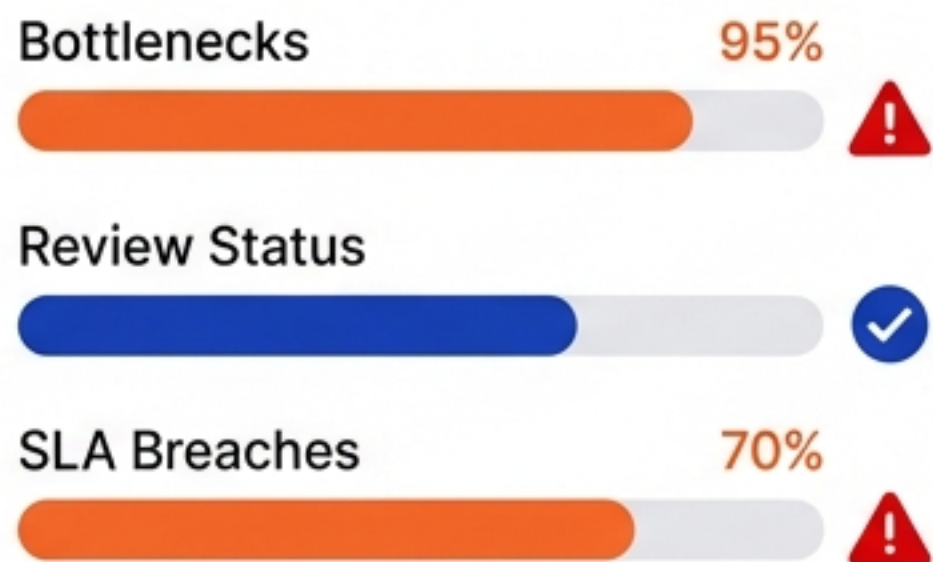
Reporting, Intelligence & Defensibility

Executive View



Portfolio Health: Outcomes, ROI, Impact

Operational View



SLA & Queues: Bottlenecks, Review Status

Audit View

[2024-10-27 09:45:02] USER_ID:104
ACTION:SUBMISSION_REVIEWED
STATUS:APPROVED
[2024-10-27 10:12:15] USER_ID:108
ACTION:EVIDENCE_UPLOADED TYPE:PDF
[2024-10-27 10:30:45]
SYSTEM:AUDIT_LOG_ENTRY
EVENT:DECISION_MADE DECISION:ACCEPT
[2024-10-27 11:05:30] USER_ID:102
ACTION:COMMENT_ADDED
COMMENT:"Looks good."

Evidence Ledger:
Completeness, Decision History

Insight Governance: Insights must show: Observation + Impact + Suggested Action. Innovation intelligence supports decisions; it does not replace them.

Why This Architecture Wins

Scalable operations for NOVATE. Defensible independence for Clients.

Value for CLIENT



Brand Independence: Their logo, their terminology.



Rigorous Compliance: NFL-grade security and auditability.



Flexible Integration: Fits into existing multi-cloud worlds.

Value for NOVATE



Operational Scale: Manage many ecosystems from one console.



IP Protection: Playbooks remain central and protected.



Central Visibility: Portfolio insights without crossing data boundaries.

The Result: Innovation that is standardized, secure, and scalable.

Appendix: Glossary & Key Concepts

Control Plane	The multi-tenant, NOVATE-operated layer containing logic, policy, and workflow.
Tenant Pod	The isolated, client-specific environment containing data, encryption keys, and evidence.
CMEK	Customer-Managed Encryption Keys (client holds the keys to their data).
NIN	NOVATE Innovation Network standards (the workflow methodology).
Playbook	NOVATE IP providing the template logic for innovation challenges.

Core Definition: NOVATE operates a single innovation control plane in GCP. Each client has an isolated ecosystem tenant that we run on their behalf.